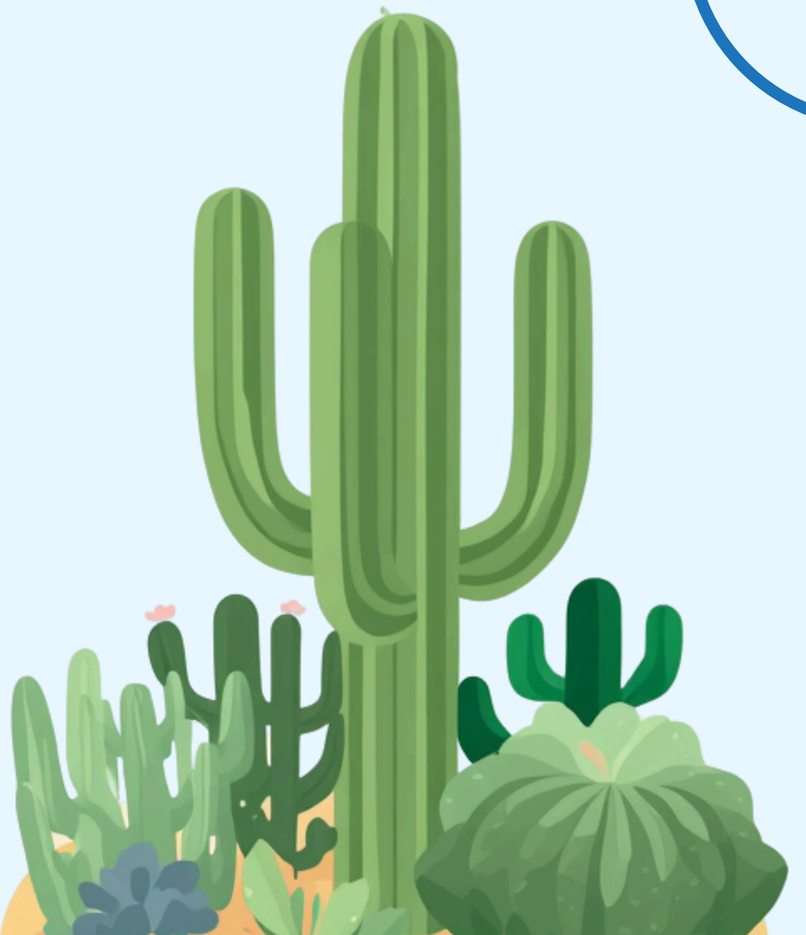
THE AMAZING CACTUS

How a Cactus Thrives in the Desert

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WHERE DOES THE CACTUS GET ENERGY?

Energy Sources: Sunlight powers all cactus processes. Carbon dioxide is absorbed from the air, water is stored in the thick stem, and minerals are drawn from the soil. Together, these raw materials fuel photosynthesis inside the cactus cells.

Energy Flow: Photosynthesis converts sunlight + CO_2 + water → Glucose + Oxygen. Glucose is then broken down through Cellular Respiration → ATP. ATP powers growth, repair, nutrient transport, and survival in the desert heat.

ENZYMES: THE PLANT'S TINY WORKERS



Enzymes speed up chemical reactions essential for life. Key enzymes include:

- Rubisco – captures CO_2 for photosynthesis
- ATP synthase – produces ATP from energy
- Glycolysis enzymes – break glucose into pyruvate
- Citric acid cycle enzymes – release stored energy

Without enzymes, life processes slow dramatically:

- Photosynthesis slows significantly
- ATP production drops
- Growth and repair decline
- The plant may wither and die

Enzymes are biological catalysts — essential for cactus survival.

CELLULAR RESPIRATION: MAKING ATP

Glycolysis: Glucose is broken down into pyruvate in the cytoplasm. This first step requires no oxygen and produces a small amount of ATP and NADH, setting the stage for further energy extraction.

Citric Acid Cycle: Pyruvate enters the mitochondria and is processed through the Citric Acid Cycle. This stage releases carbon dioxide and produces electron carriers NADH and $FADH_2$, which store energy for the next step.

Electron Transport Chain: NADH and $FADH_2$ donate electrons to the chain, driving ATP synthase to produce the majority of ATP. Final products are ATP, water, and CO_2 — powering growth, repair, and survival.

SPECIAL DESERT ADAPTATION: CAM PHOTOSYNTHESIS



At night, stomata open to absorb CO_2 from the air. CO_2 is converted into malic acid and stored in vacuoles, allowing the cactus to gather carbon without losing water through evaporation.

During the day, stomata remain closed to prevent water loss. Stored CO_2 is released from malic acid and used with sunlight to drive photosynthesis, producing glucose for energy and survival.

WHY ATP MATTERS

ATP supports the essential functions that keep the cactus alive and thriving:
- Growth of stems, roots, and spines
- Repair of damaged cells after stress
- Transport of minerals and nutrients throughout the plant
- Protein synthesis for building enzymes and structures
- Flower and seed production for reproduction

Without ATP, the cactus cannot grow, repair itself, or survive. ATP is the rechargeable energy currency that powers every cellular activity — from the smallest enzyme reaction to the largest structural growth. All energy from photosynthesis and cellular respiration ultimately flows into ATP production.

KEY TAKEAWAYS







- Cactus uses sunlight to produce glucose through CAM photosynthesis
- Enzymes speed up both photosynthesis and cellular respiration
- Cellular respiration converts glucose into ATP energy

- CAM photosynthesis conserves water by closing stomata during the day
- ATP powers all plant activities: growth, repair, transport, and survival

REFERENCES

OpenStax Biology 2e. (2018). Biology 2e. OpenStax. https://openstax.org/details/books/biology-2e

Campbell, N. A., & Reece, J. B. (2021). Campbell Biology (12th ed.). Pearson.

University of the People. (2026). BIOL 1301 Unit 3 Learning Materials. University of the People.

